nf-core

A community effort to collect curated nextflow pipelines

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NATIONAL GENOMICS INFRASTRUCTURE

A long time ago in a institute far, far away....

NEXT

THE 2017 HACKATHON

FLOW



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Standardising Swedish genomics analyses using nextflow

Phil Ewels

@ewels

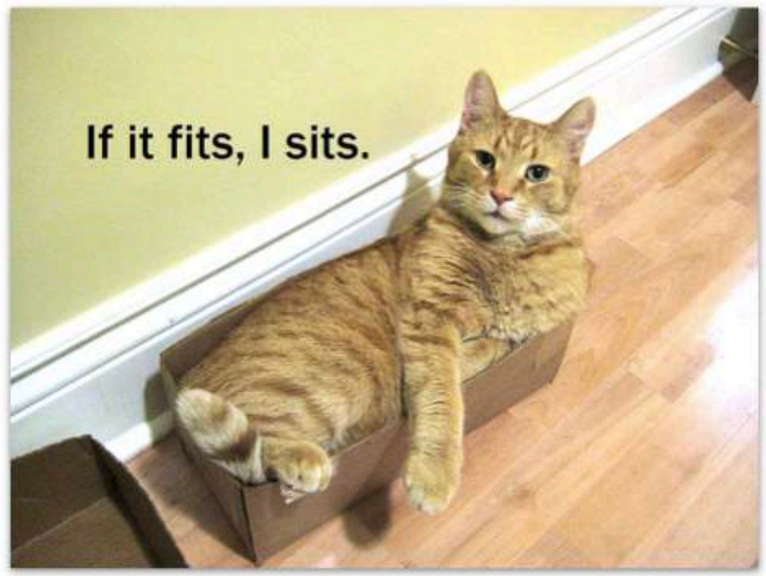
@tallphil

Nextflow Meeting

2017-09-14

CRG, Barcelona

use containers





NGI pipelines

NGI-RNAseq

CAW

Cancer Analysis Workflow

NGI-MethylSeq

NGI-ExoSeq

NGI-smRNAseq

NGI-RNAfusion

NGI-ChIPseq

NouGAT *(de-novo)*



Alex Peltzer

QBiC





Andreas Wilm

A*STAR Genome Institute of Singapore



Open source

Fully featured

Consistent usage

Excellent documentation



What we'd really like is...





nf-core



User community



Workflow collection



Helper tools

nf-core

https://github.com/**nf-core/**

https://nf-co.re/

https://twitter.com/**nf_core/**

https://hub.docker.com/u/**nfcore/**

nf-core

38
team members

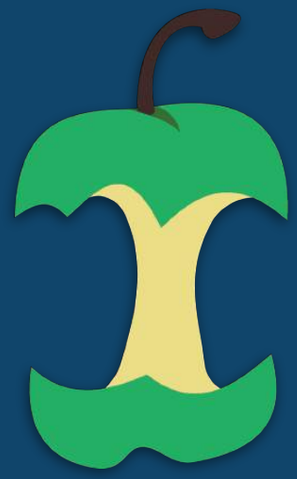
12+
institutions

1440
unique website visitors

322
twitter followers



nf-core



CRG Centre for Genomic Regulation

NATIONAL GENOMICS INFRASTRUCTURE

CANCER RESEARCH UK | BEATSON INSTITUTE

QBiC

SciLifeLab

lifebit

A*STAR Genome Institute of Singapore

SLU

University of Colorado Boulder

SYSUCC SINCE 1964

中山大學 肿瘤防治中心 SUN YAT-SEN UNIVERSITY CANCER CENTER

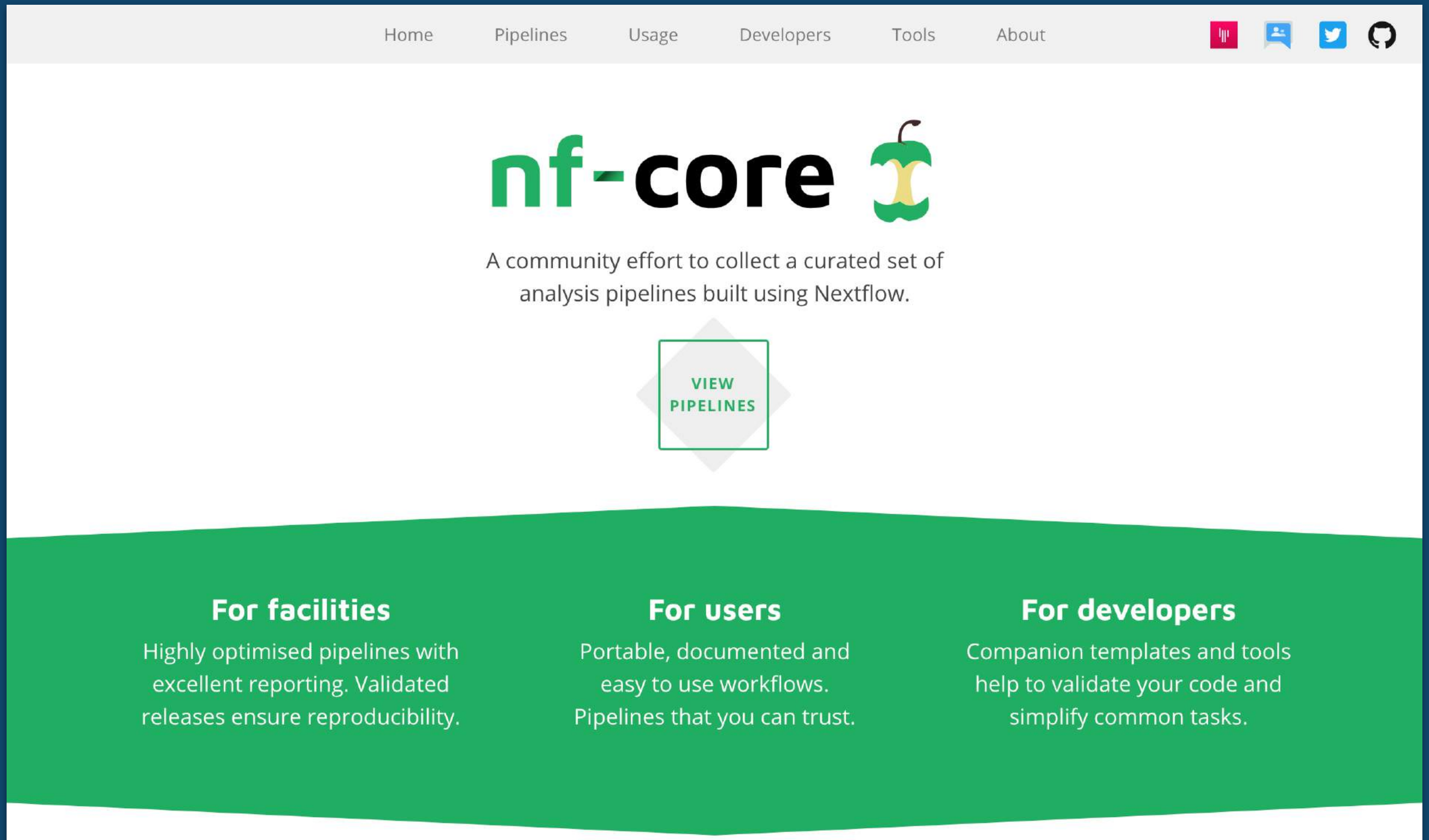
International Agency for Research on Cancer

World Health Organization

THE FRANCIS CRICK INSTITUTE

wellcome sanger institute

https://nf-co.re/



https://nf-co.re/

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Pipelines

Browse the **16** pipelines that are currently available as part of nf-core.

Available Pipelines

Can you think of another pipeline that would fit in well? Let us know!

Search keywords | Filter: Released 4 | Under development 12 | Sort: Last Release | Alphabetical | Status | Stars

nf-core/eager ✓ ☆ 8

adna ancientdna pathogen-genomics population-genetics

A fully reproducible and state of the art ancient DNA analysis pipeline.

Version **2.0.2** Published 3 days ago

nf-core/rnaseq ✓ ☆ 48

rna rna-seq

RNA sequencing analysis pipeline using STAR or HISAT2, with gene counts and quality control

Version **1.1** Published 1 month ago

nf-core/hlatyping ✓

dna hla hla-typing immunology optitype personalized-medicine rna

Precision HLA typing from next-generation sequencing data

Version **1.1.1** Published 3 months ago

nf-core/methylseq ✓ ☆ 15

bisulfite-sequencing dna-methylation methyl-seq

Methylation (Bisulfite-Sequencing) analysis pipeline using Bismark or bwa-meth + MethylDackel

Version **1.1** Published 3 months ago

https://nf-co.re/

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Tools

A companion tool is available for nf-core to help with common tasks.

Table of contents

- Installation
- Listing pipelines (`nf-core list`)
- Downloading pipelines for offline use (`nf-core download`)
- Listing software licences: List software licences for a given workflow (`nf-core licences`)
- Creating a new workflow (`nf-core create`)
- Checking a pipeline against nf-core guidelines (`nf-core lint`)
- Bumping a pipeline version number (`nf-core bump-version`)

Installation

You can install `nf-core/tools` from PyPI using pip as follows:

```
pip install nf-core
```

If you would like the development version instead, the command is:

```
pip install --upgrade --force-reinstall git+https://github.com/nf-core/tools.git
```

ADDING A PIPELINE

GUIDELINES

- Nextflow
- MIT
- Documentation
- Responsible person
- Docker
- CI Testing
- Stable releases
- Passing tests
- Bioconda*

```
(work)  ~/demo » nf-core create
```



```
Workflow Name: mynewpipeline
Description: This is a new workflow that is going to be really amazing.
Author: Phil Ewels

INFO: Creating new nf-core pipeline: nf-core/mynewpipeline

INFO: Initialising pipeline git repository

INFO: Done. Remember to add a remote and push to GitHub:
  cd /Users/ewels/demo/nf-core-mynewpipeline
  git remote add origin git@github.com:USERNAME/REPO_NAME.git
  git push
(work)  ~/demo »
```

```
(work)  ~/demo » cd nf-core-mynewpipeline/
(work)  ~/.../nf-core-mynewpipeline (master) » ls -l
total 96
-rw-r--r--  1 ewels  staff    153 20 Nov 11:30 CHANGELOG.md
-rw-r--r--  1 ewels  staff   3250 20 Nov 11:30 CODE_OF_CONDUCT.md
-rw-r--r--  1 ewels  staff    289 20 Nov 11:30 Dockerfile
-rw-r--r--  1 ewels  staff   1062 20 Nov 11:30 LICENSE
-rw-r--r--  1 ewels  staff   1595 20 Nov 11:30 README.md
-rw-r--r--  1 ewels  staff    410 20 Nov 11:30 Singularity
drwxr-xr-x  5 ewels  staff    160 20 Nov 11:30 assets/
drwxr-xr-x  4 ewels  staff    128 20 Nov 11:30 bin/
drwxr-xr-x  7 ewels  staff    224 20 Nov 11:30 conf/
drwxr-xr-x  8 ewels  staff    256 20 Nov 11:30 docs/
-rw-r--r--  1 ewels  staff    193 20 Nov 11:30 environment.yml
-rw-r--r--  1 ewels  staff  13239 20 Nov 11:30 main.nf
-rw-r--r--  1 ewels  staff   3500 20 Nov 11:30 nextflow.config
(work)  ~/.../nf-core-mynewpipeline (master) »
```

```
(work)  ~/.../nf-core-mynewpipeline (master) » git log
commit 20ad7158bc395dfcc704d5546da6444b043982f4 (HEAD -> master)
Author: Phil Ewels <phil.ewels@scilifelab.se>
Date:   Tue Nov 20 11:30:34 2018 +0100

    initial template build from nf-core/tools, version 1.3.dev0
(work)  ~/.../nf-core-mynewpipeline (master) »
```

```
    // TODO nf-core: Add in log files from your new processes for MultiQC to find!
    file ('fastqc/*') from fastqc_results.collect().ifEmpty([])
    file ('software_versions/*') from software_versions_yaml
    file workflow_summary from create_workflow_summary(summary)

    output:
    file "*multiqc_report.html" into multiqc_report
    file "*_data"

    script:
    rtitle = custom_runName ? "--title \"$custom_runName\"" : ''
    rfilename = custom_runName ? "--filename " + custom_runName.replaceAll('\\W','_
_multiqc_report" : ''
    // TODO nf-core: Specify which MultiQC modules to use with -m for a faster run
    """
    multiqc -f $rtitle $rfilename --config $multiqc_config .
    """
}


/*
 * STEP 3 - Output Description HTML
 */
process output_documentation {
    publishDir "${params.outdir}/Documentation", mode: 'copy'

    input:
:
```

```
(work)  ~/GitHub/.../methylseq (master) » nf-core lint .

                                          ,--./,-.
          ___     __   __   __   ___     /,-._.--~\
    |\ | |__  __ /  ` /  \ |__) |__         }  {
    | \| |       \__, \__/ |  \ |___     \`-._,-`-,
                                          `._,._,'

Running pipeline tests  [####################################]  100%  None

INFO: ==========
 LINTING RESULTS
================
  80 tests passed   1 tests had warnings   0 tests failed

WARNING: Test Warnings:
  http://nf-co.re/errors#8: Conda package is not latest available: picard=2.18.14, 2.18.16 available
(work)  ~/GitHub/.../methylseq (master) »
```

```
(work)  ~/GitHub/.../methylseq (master) » nf-core lint .

                                          ,--./,-.
          ___     __   __   __   ___     /,-._.--~\
    |\ | |__  __ /  ` /  \ |__) |__         }  {
    | \| |       \__, \__/ |  \ |___     \`-._,-`-,
                                          `._,._,'

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  80 tests passed   1 tests had warnings   0 tests failed

WARNING: Test Warnings:
  http://nf-co.re/errors#8: Conda package is not latest available: picard=2.18.14, 2.18.16 available
(work)  ~/GitHub/.../methylseq (master) »
```

- Tags and $TRAVIS_TAG must all match one another

Error #8 - Conda environment tests

> These tests only run when your pipeline has a root file called `environment.yml`

- The environment `name` must match the pipeline name and version
  - The pipeline name is defined in the config variable `manifest.name`
  - Replace the slash with a hyphen as environment names shouldn't contain that character
  - Example: For `nf-core/test` version 1.4, the conda environment name should be `nf-core-test-1.4`

Each dependency is checked using the Anaconda API service. Dependency sublists are ignored with the exception of `- pip`: these packages are also checked for pinned version numbers and checked using the PyPI JSON API.

Note that conda dependencies with pinned channels (eg. `conda-forge::openjdk`) are fine and should be handled by the linting properly.

Each dependency can have the following lint failures and warnings:

- (Test failure) Dependency does not have a pinned version number, eg. `toolname=1.6.8`
- (Test failure) The package cannot be found on any of the listed conda channels (or PyPI if `pip`)
- (Test warning) A newer version of the package is available

Error #9 - Dockerfile for use with Conda environments

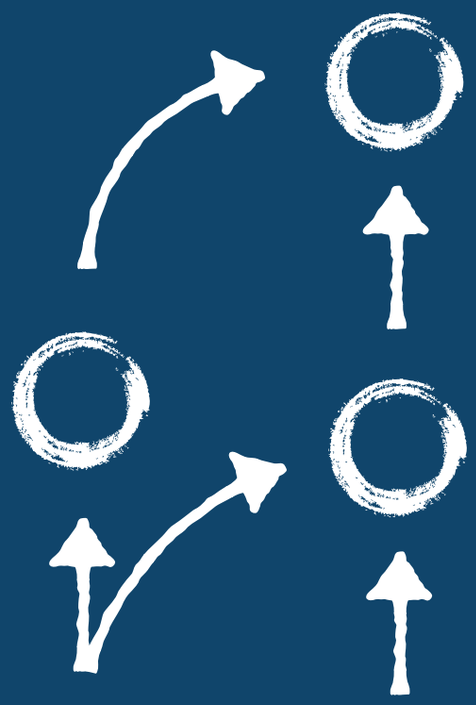
> This test only runs if there is both `environment.yml` and `Dockerfile` present in the workflow.

If a workflow has a conda `environment.yml` file (see above), the `Dockerfile` should use this to create the container. Such `Dockerfile`s can usually be very short, eg:

```
 FROM nfcore/base
LABEL authors="your@email.com" \
      description="Container image containing all requirements for nf-core/EXAMPLE pipeline"

COPY environment.yml /
RUN conda env update -n root -f /environment.yml && conda clean -a
```

SYNCHRONISATION





Sven Fillinger

Important pipeline nf-core update! (version 1.3) #39

Edit

Open **nf-core-bot** wants to merge 5 commits into `dev` from `TEMPLATE`

Conversation 0 | Commits 5 | Checks 0 | Files changed 36 | +691 −424

nf-core-bot commented 2 minutes ago — Member

Some important changes have been made in the nf-core pipelines templates.
Please make sure to merge this in ASAP and make a new minor release of your pipeline.

Follow the link nf-core/tools

sven1103 and others added some commits on 7 Aug

Test the template	✗ 8f3dd6b
Update nf-core pipeline template.	✗ 4df38af
Update nf-core pipeline template.	● af51711
Update nf-core pipeline template.	● feb5d4a
Update nf-core pipeline template.	● 3c54be1

Add more commits by pushing to the **TEMPLATE** branch on **nf-core/hlatyping**.

Reviewers
No reviews

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Notifications

36 .travis.yml

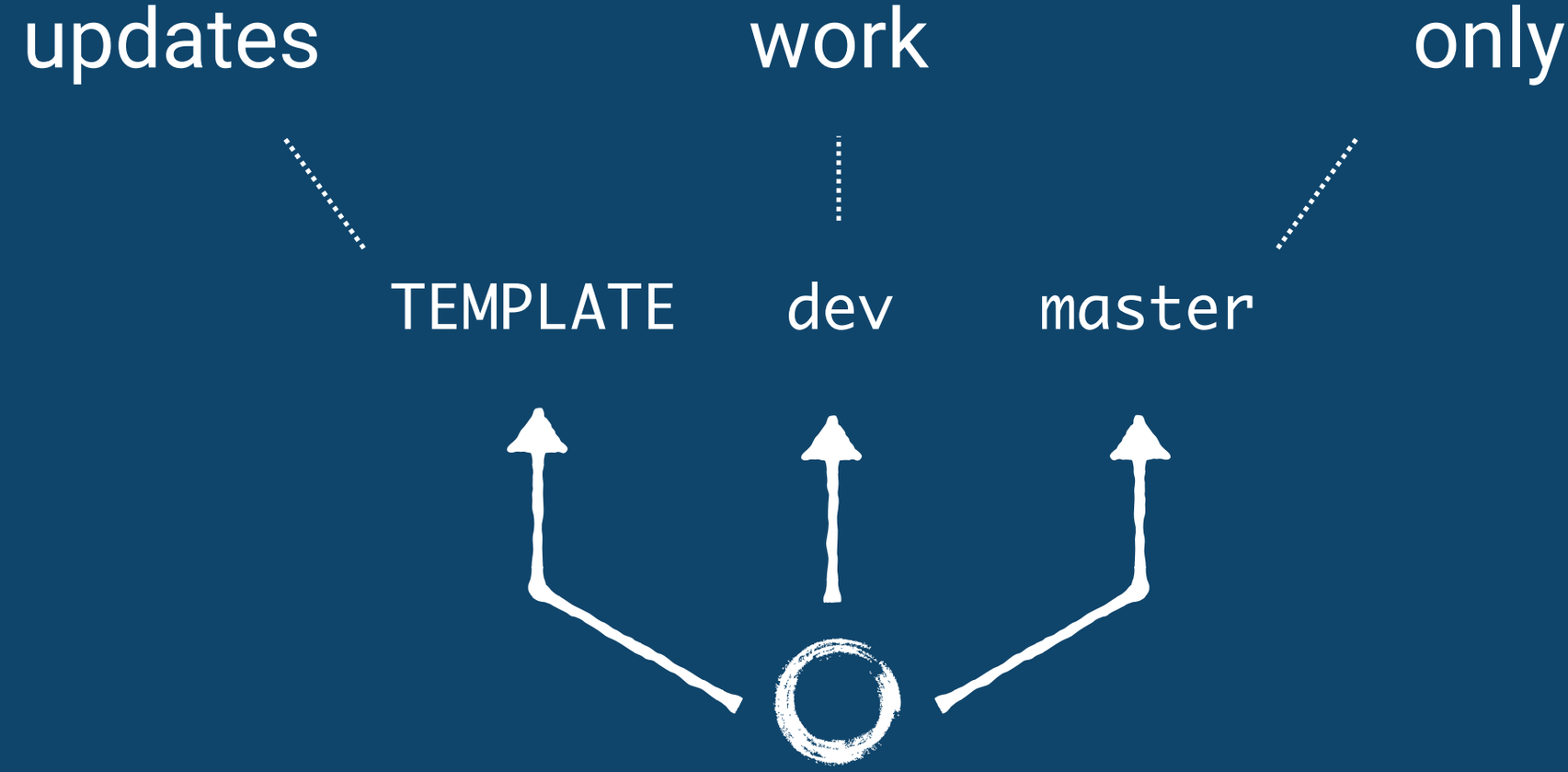
Copy path | View file

```
...  ...  @@ -1,31 +1,37 @@
1    1      sudo: required
2         - language: java
     2    + language: python
3    3      jdk: openjdk8
4         - services:
5         -   - docker
6         - python:
7         -   - "2.7"
     4    + services: docker
     5    + python: '3.6'
8    6      cache: pip
9    7      matrix:
10   8        fast_finish: true
11   9
     10   + before_install:
     11   +   # PRs to master are only ok if coming from dev branch
     12   +   - '[ $TRAVIS_PULL_REQUEST = "false" ] || [ $TRAVIS_BRANCH != "master" ] || ([ $TRAVIS_PULL_REQUEST_SLUG = $TRAVIS_RE
     13   +   # Pull the docker image first so the test doesn't wait for this
     14   +   - docker pull nfcore/hlatyping
     15   +   # Fake the tag locally so that the pipeline runs properly
     16   +   - docker tag nfcore/hlatyping nfcore/hlatyping:latest
     17   +
12   18     install:
13   19       # Install Nextflow
14        -   - mkdir /tmp/nextflow
15        -   - cd /tmp/nextflow
```

Automated template updates

All development work

Latest release only

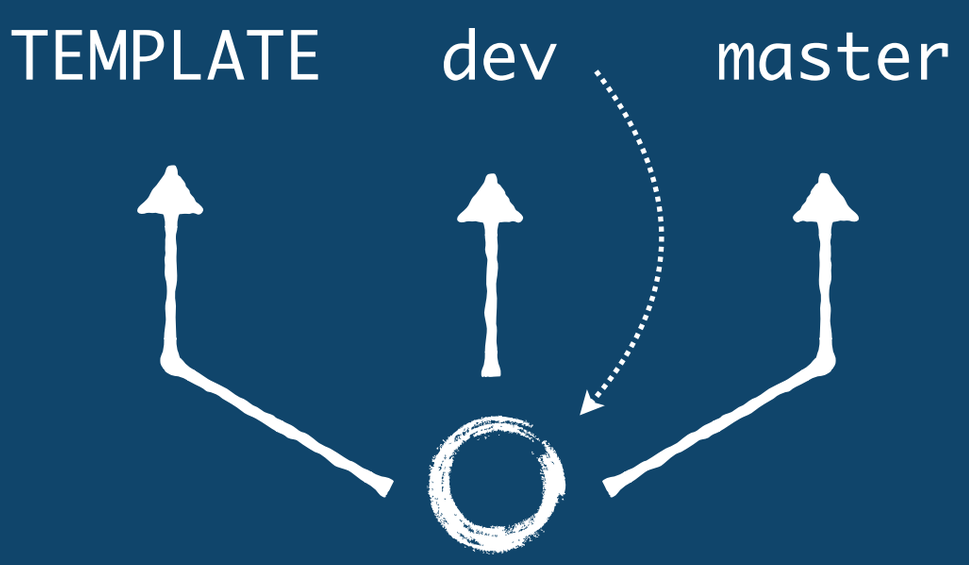


Initial release of nf-core/deepvariant! 🎉 Latest release

Full pipeline review - pseudo PR ❌ WIP 156 😱

#3 by ewels was closed a day ago • Approved

Group review of first release



https://nf-co.re/

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Adding a new pipeline

Follow this walkthrough to add a new pipeline to nf-core.

Before you start

So, you want to add a new pipeline to nf-core - brilliant! Before you start typing, check that you're happy with the following points:

- You're familiar with nf-core and nextflow (see our introduction docs).
- You're used to working with `git` and GitHub (see a nice tutorial here)
- The workflow you're thinking of meets the nf-core guidelines.

Join the community

At its heart, nf-core is a community - to add a pipeline you need to be part of that community! Please request to join the nf-core GitHub organisation) and introduce yourself on gitter or the mailing list.

It's good to introduce your idea early on so that it can be discussed before you spend lots of time coding.

Create a pipeline from the template

The guidelines for nf-core pipelines are pretty strict, but if you start your pipeline by using the nf-core template (`nf-core create` - see the docs) then your life will be much easier. This tool does lots of things for you: it gives you the correct file structure and boiler plate

GETTING INVOLVED

GETTING INVOLVED



https://nf-co.re/



Email list GITTER slack



https://github.com/nf-core/



Hackathons!

GETTING INVOLVED



Hackathons!

August 6-10 (2018) Stockholm

November 22-23 (2018) Barcelona

April 8-10 (2019) Tübingen

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Icons: flaticon.com

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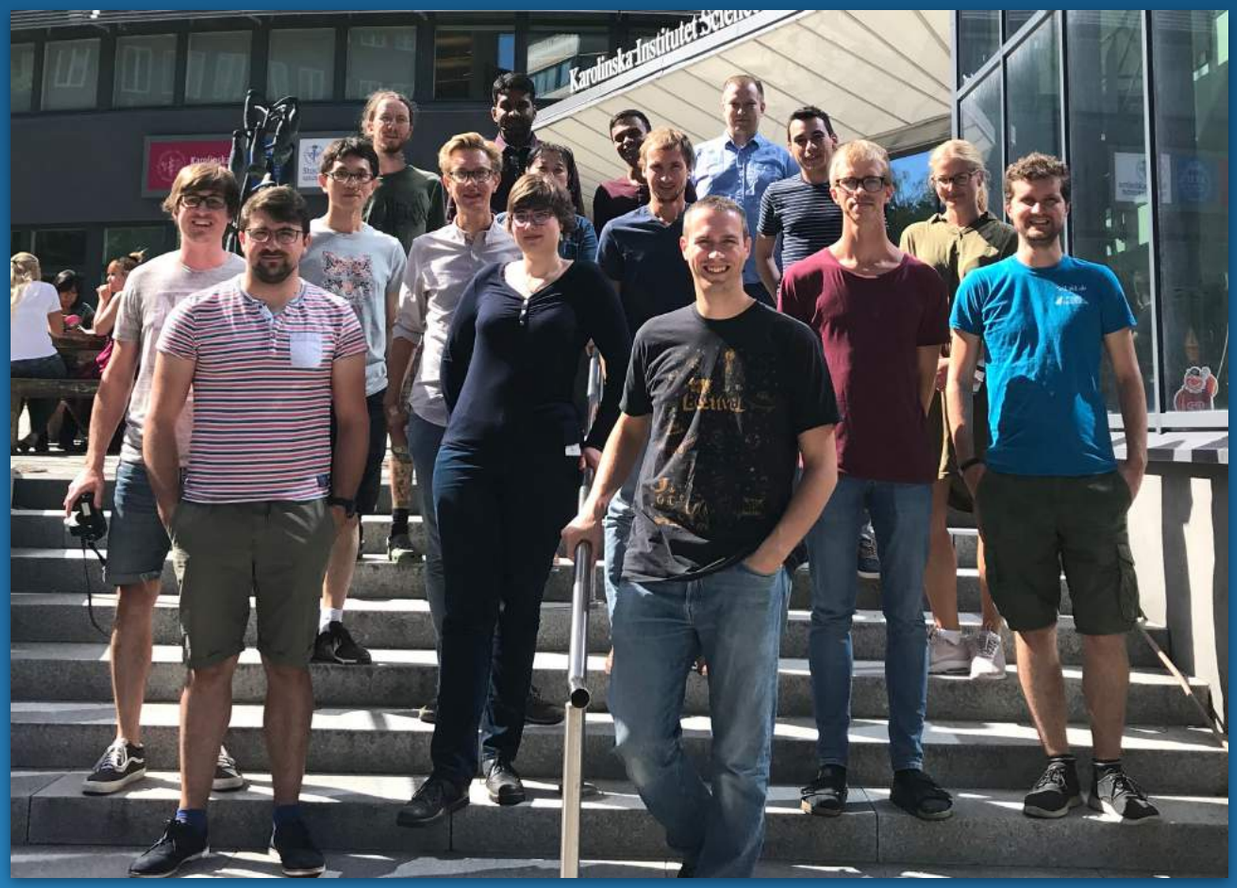
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MultiQC

Sarek

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We're hiring!

Systems / devops

No biology background required

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